Introduction to Photogrammetry

Lecture 04

Classification of Sensing Devices

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In Today’s Class

1. Sensing Devices in Photogrammetry
2. Radiometric Energy and the Electromagnetic Spectrum
3. Classification of Sensing Devices
    1. Active Sensing Systems
    2. Passive Sensing Systems

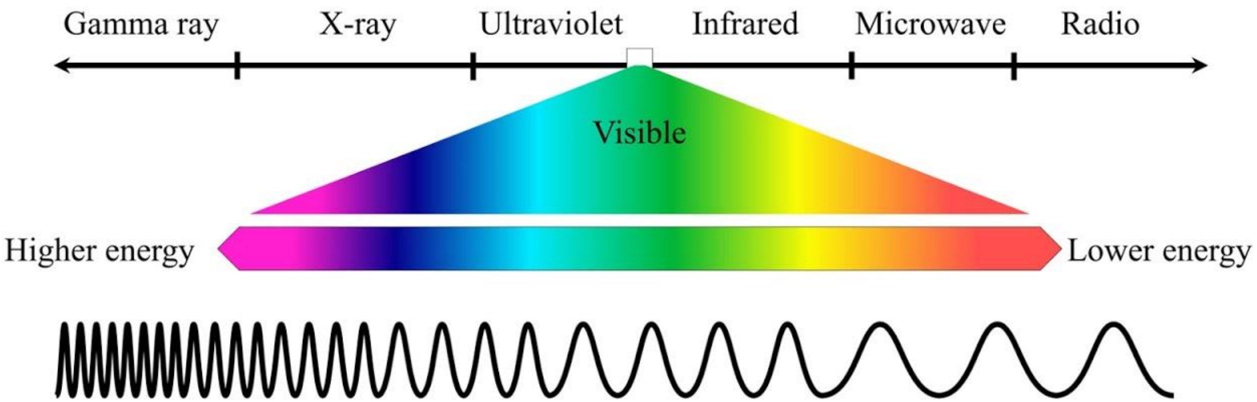
1. Sensing Devices in Photogrammetry

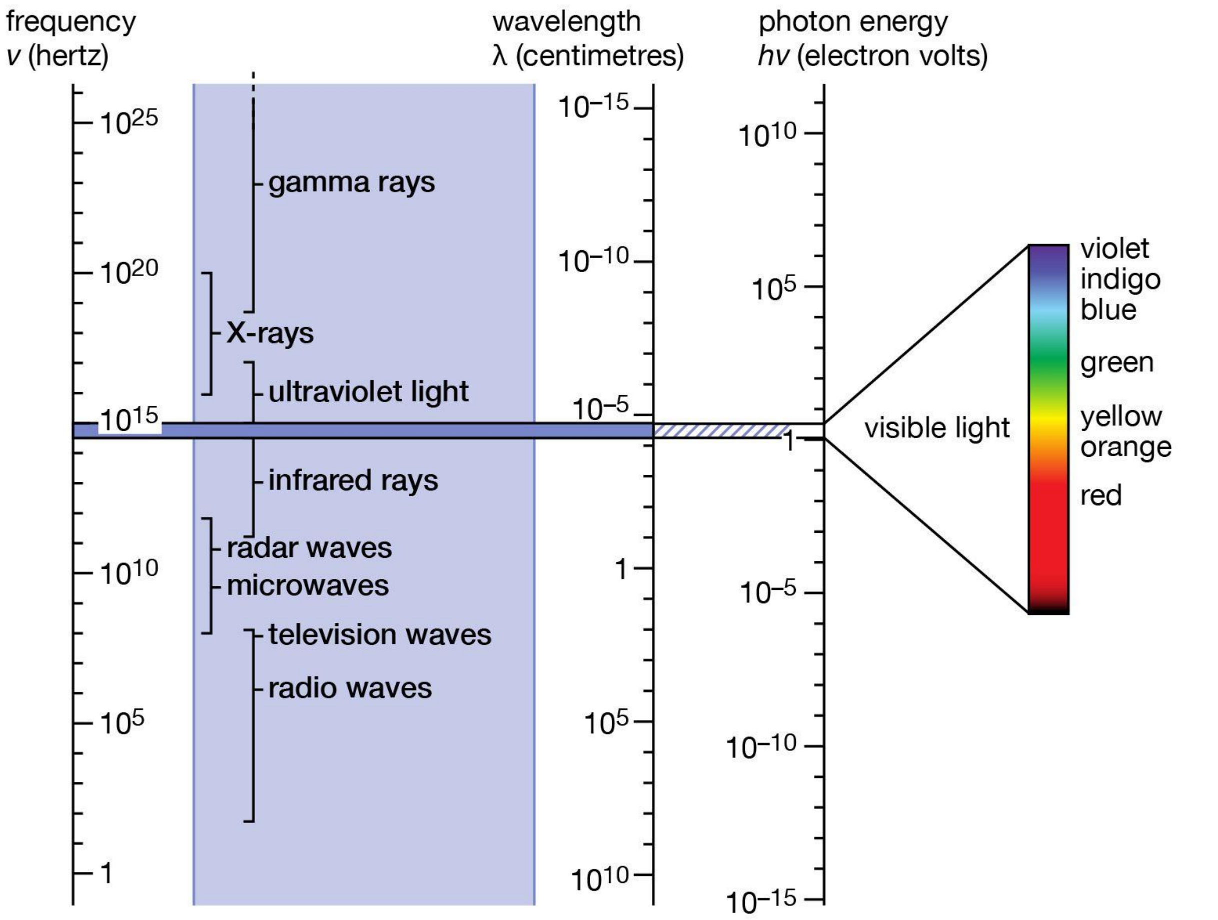
- In photogrammetry and remote sensing, a sensing device is an instrument that detects and records radiometric energy from objects on the Earth's surface.
- Radiometric energy refers to electromagnetic energy that is reflected or emitted by objects.
- The recorded energy is converted into data that can later be processed into images, maps, or measurements.
- Different sensing devices are used depending on the method of energy detection and the type of data required.
- Understanding the classification of sensing devices is important for selecting appropriate technologies for mapping and spatial analysis.

2. Radiometric Energy and the Electromagnetic Spectrum

- Sensing devices operate by detecting electromagnetic radiation reflected or emitted by objects.
- Electromagnetic radiation includes a wide range of wavelengths known as the electromagnetic spectrum.
- Important portions of the spectrum used in photogrammetry and remote sensing include visible light, infrared radiation, and microwave radiation.
- Different sensors are designed to detect specific portions of the electromagnetic spectrum.
- The choice of wavelength influences the type of information that can be obtained from the Earth's surface.

Electromagnetic Spectrum





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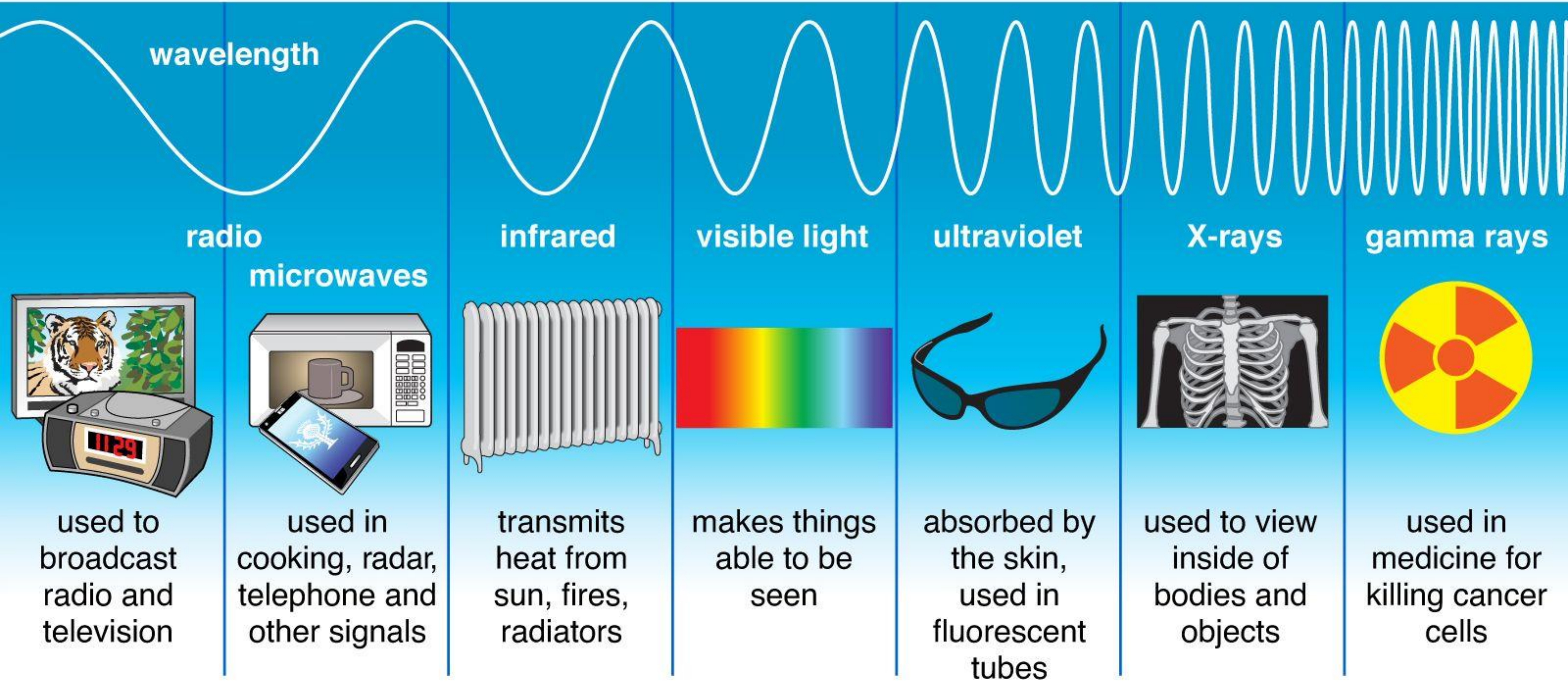
Types of Electromagnetic Radiation



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3. Classification of Sensing Devices

- Sensing devices used in photogrammetry can be classified based on how they obtain and record energy.
- The two main categories are active sensing systems and passive sensing systems.
- **Active sensing systems** generate their own energy and then measure the reflected signal from the Earth's surface.
- **Passive sensing systems** do not generate energy but instead record naturally available energy reflected or emitted from objects.
- Both active and passive systems are widely used in Earth observation and mapping applications.

3.1 Active Sensing Systems

- Active sensing systems emit their own electromagnetic energy toward the Earth's surface.
- After the energy interacts with the surface, part of it is reflected back to the sensor.
- The sensor records the reflected signal and uses it to generate images or measurements.
- Because these systems provide their own energy source, they can operate independently of sunlight.
- Active sensors can therefore function during nighttime or under low illumination conditions.

Example of Active Sensing: Radar

- Radar (Radio Detection and Ranging) is a common example of an active sensing system.
- Radar sensors transmit microwave signals toward the ground surface.
- The transmitted energy interacts with objects and terrain features and is partially reflected back to the sensor.
- The reflected signals are analyzed to determine the location, distance, and characteristics of objects.
- Radar systems are widely used in remote sensing, navigation, and weather monitoring.
- An important advantage of radar systems is their ability to penetrate clouds, haze, and certain atmospheric conditions.

Side Looking Airborne Radar (SLAR)

- An operational radar system sometimes used for photogrammetric applications is called Side Looking Airborne Radar (SLAR).
- In this system, an antenna mounted on the belly of an aircraft transmits microwave energy sideways, perpendicular to the direction of flight.
- The transmitted microwave energy travels toward the ground surface where it is scattered and partially reflected.
- A portion of this reflected energy returns to the same antenna that transmitted the signal.
- The time interval between the transmitted signal and the received signal is measured to determine the distance between the radar antenna and the ground surface.

3.2 Passive Sensing Systems

- Passive sensing systems do not generate their own energy.
- Instead, they record electromagnetic radiation that is naturally available in the environment.
- Most passive sensors depend on sunlight reflected from the Earth's surface.
- Some passive sensors also detect thermal radiation emitted by objects.
- Passive sensing systems are commonly used in aerial photography and satellite imaging.

Categories of Passive Sensing Systems

- Passive sensing systems can be further classified into two main types.
- The first type is image forming systems, which produce images of the Earth's surface.
- The second type is spectral data systems, which record detailed spectral information about objects.
- Both types of sensors are used to analyze physical and environmental characteristics of the Earth's surface.
- The choice between these systems depends on the specific objectives of the remote sensing study.

Image Forming Systems

- Image forming systems produce images that represent the spatial distribution of reflected or emitted energy.
- These systems establish both geometric relationships and radiometric relationships between the object space and the image space.
- The recorded radiation at each pixel corresponds to a specific location on the ground.
- Image forming systems are widely used in mapping, photogrammetry, and spatial analysis.
- These systems allow analysts to interpret terrain features, land cover, and environmental patterns.

Framing Systems

- Framing systems capture the entire image at a single instant in time.
- In this method, the complete photographic frame is exposed simultaneously.
- Because all parts of the image are recorded at the same moment, geometric relationships within the image are preserved.
- Traditional aerial photographic cameras used in photogrammetry are examples of framing systems.
- Framing systems are particularly suitable for mapping and photogrammetric measurements.

Scanning Systems

- Scanning systems record image data sequentially rather than capturing the entire image at once.
- These systems collect data line by line, which is commonly referred to as scanline acquisition.
- As the sensor moves across the ground, successive scanlines are recorded and combined to form a complete image.
- Scanning systems are widely used in satellite remote sensing platforms.
- They are especially useful for collecting multispectral and hyperspectral data.

Spectral Data Systems

- Spectral data systems are sensors designed to measure the spectral characteristics of objects.
- These systems record the intensity of radiation in different wavelength bands.
- The collected spectral information helps identify materials and surface properties.
- Spectral sensors are commonly used for vegetation analysis, mineral exploration, and environmental monitoring.
- Examples include multispectral scanners and hyperspectral sensors.

Thank you!

Any Questions